

Hybrid CNN-LSTM and Physics-Informed Architectures for Real-Time Tumor Ablation Monitoring on Edge Devices

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Abstract—This paper presents a comprehensive study on estimating temperature dynamics from ultrasonic video sequences using deep learning models tailored for edge deployment, specifically for monitoring temperature during heat treatment of tumor tissue. We propose a specialized CNN-LSTM architecture for this task and compare it against Pretrained ResNets and Physics-Informed Neural Networks (PINNs) in terms of accuracy and computational efficiency. To enhance robustness and provide a transparent monitoring system, we integrate Uncertainty Quantification (UQ) via Deep Ensembles and Bayesian Neural Networks. Furthermore, we propose a lightweight preprocessing pipeline incorporating dense optical flow to capture temporal dynamics explicitly. We benchmark these models on a simulated edge environment, demonstrating that our optimized architectures achieve real-time performance (≥ 30 FPS) on low-cost, resource-constrained devices like the Raspberry Pi 4 and Jetson Nano, while maintaining high predictive accuracy.

Index Terms—Video Regression, Physics-Informed Machine Learning, Edge Computing, Bayesian Neural Networks, Optical Flow, Thermal Ablation, Medical Imaging

I. INTRODUCTION

Estimating physical parameters, such as temperature, from medical imaging data is a critical challenge in non-invasive thermal ablation therapies. In procedures like High-Intensity Focused Ultrasound (HIFU), precise temperature monitoring is essential to ensure tumor destruction while preserving healthy surrounding tissue. However, direct temperature measurement is often impossible or requires invasive probes. In this work, we explore the feasibility of using ultrasonic video sequences to regress temperature values, leveraging temporal coherence and visual cues present in the footage.

A major constraint in clinical deployment is data privacy. Transmitting sensitive patient data to cloud servers for processing poses significant security risks and regulatory hurdles. Therefore, inference must be performed locally on edge devices. Furthermore, to ensure widespread accessibility, these devices must be cost-effective and easily retrofittable to existing ultrasonic machines. This necessitates models that are not only accurate but also lightweight enough to run on low-cost hardware like the Raspberry Pi or Jetson Nano.

Finally, for clinical adoption, black-box predictions are insufficient. We emphasize the need for a transparent monitoring system by integrating Uncertainty Quantification (UQ), providing clinicians with confidence intervals alongside temperature estimates.

We address these challenges by:

- Proposing a custom, lightweight CNN-LSTM architecture optimized for ultrasonic video regression.
- Integrating dense optical flow as an explicit input feature.
- Benchmarking inference latency via ONNX Runtime on CPU.

II. METHODOLOGY

III. AUTOMATED SENSOR LOCALIZATION

To bridge the gap between the sparse temperature readings and the dense visual data, we developed an automated method to localize the four temperature sensors within the video frames. The sensors appear as bright, circular regions of high intensity. However, simply thresholding often yields false positives due to reflections or noise. We therefore enforce a geometric constraint based on the known physical layout of the sensors.

A. Methodology

Our robust localization pipeline consists of the following steps:

- 1) **Dynamic ROI Cropping (Active Zone):** Before detection, we identify the relevant "Active Zone" to eliminate high-intensity artifacts outside the region of interest. We compute the vertical projection of the frame intensity (row-wise mean) to find the peak row y_{peak} . The image is then cropped to the region $[y_{peak} - 180, y_{peak} + 180]$. This dynamic cropping accounts for vertical shifts in the setup across different experiments while preserving the sensors.
- 2) **Preprocessing and Enhancement:** We average the first 15 frames of the video sequence to suppress transient noise. To assist detection in varying lighting conditions, we apply Contrast Limited Adaptive Histogram

Equalization (CLAHE) to the cropped grayscale image, followed by Gaussian blurring.

- 3) **Candidate Detection:** We apply a binary threshold to the enhanced image and extract contours. Candidates are filtered based on geometric properties typical of the sensors:

- Area constraint: $20 < \text{Area} < 5000$ pixels.
- Radius constraint: $r < 60$ pixels.

- 4) **Pattern Optimization (Axis-Aligned):** Let $C = \{c_1, c_2, \dots, c_N\}$ be the set of valid candidate centroids. We seek the subset $S \subset C$ of size 4 that best approximates an *axis-aligned* rectangle. We iterate through all combinations of 4 candidates and minimize a cost function that penalizes:

$$\text{Cost} = \Delta_{\text{width}} + \Delta_{\text{height}} + \Delta_{\text{diag}} + 2 \cdot \text{AlignmentPenalty} \quad (1)$$

where Δ represents the normalized difference between opposite sides and diagonals, and AlignmentPenalty penalizes deviations from horizontal and vertical axes. This strictly enforces the grid layout, rejecting rotated parallelograms.

- 5) **Grid Assignment:** The four selected points are mapped back to the original full-frame coordinates and sorted to assign identifiers M1–M4.

B. Verification

This combinatorial optimization approach proved significantly more robust than simple K-Means clustering or naive sorting, achieving 100% detection rate across the dataset (15/15 videos) compared to unstable results with simpler methods.

C. Data Preprocessing and Input Representation

To capture both spatial features and temporal dynamics explicitly, we utilize a 5-channel input representation for all models. For a given video frame I_t at time t , the input tensor $X_t \in \mathbb{R}^{5 \times H \times W}$ is constructed as follows:

- 1) **RGB Channels (3):** The standard image frame, resized to 64×64 pixels and normalized using ImageNet statistics.
- 2) **Optical Flow Channels (2):** We compute dense optical flow (dx, dy) between the previous frame I_{t-1} and the current frame I_t using the Farneback algorithm [1]. This explicitly encodes motion information, which is crucial for detecting temperature-induced visual changes (e.g., convection currents or heat shimmer).

This preprocessing step adds a small computational overhead (approx. 0.82 ms on CPU) but significantly enriches the input data.

D. Model Architectures

We evaluate four distinct architectures designed to balance accuracy and computational efficiency:

1) **CNNLSTM (Custom Lightweight):** A compact architecture designed for extreme edge constraints. It consists of a 3-layer CNN feature extractor (16, 32, 64 filters) followed by a Global Average Pooling layer and a single LSTM layer (hidden size 64). This model has only $\sim 60k$ parameters.

2) **Pretrained CNNLSTM:** Leverages a ResNet-18 [2] backbone pretrained on ImageNet. To accommodate the 5-channel input, we modify the first convolutional layer by extending its weights: the first 3 channels are initialized from the pretrained RGB weights, while the 2 flow channels are initialized using Kaiming initialization. The extracted features feed into an LSTM (hidden size 128).

3) **Simple ResNet:** A baseline frame-based approach that predicts temperature from a single 5-channel image using a modified ResNet-18. It lacks explicit temporal memory (LSTM) but benefits from the strong spatial feature extraction of the ResNet architecture.

4) **Physics-Informed CNNLSTM:** Identical in architecture to the Pretrained CNNLSTM but trained with a specialized loss function. The loss includes a physics regularization term based on Newton’s Law of Cooling:

$$\mathcal{L}_{\text{physics}} = \left| \frac{dT}{dt} + k(T - T_{\text{env}}) \right|^2 \quad (2)$$

This guides the model to learn physically consistent temperature dynamics, improving generalization in data-sparse regimes [3].

E. Advanced Temporal Modeling: Liquid Time Constants

While Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks are standard for time-series data, they fundamentally assume discrete time steps. However, the physical process of heat diffusion in biological tissue – described by the Bioheat Transfer Equation (BHTE) – is a continuous-time dynamical system.

$$\rho c \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q_{\text{source}} - w_b c_b (T - T_{\text{art}}) \quad (3)$$

To better align our neural architecture with this physical reality, we propose integrating Liquid Time Constant (LTC) networks [4]. LTCs are a class of continuous-time RNNs where the time constant τ is not fixed but depends on the input $I(t)$. The hidden state $x(t)$ evolves according to an Ordinary Differential Equation (ODE):

$$\frac{dx(t)}{dt} = - \left[\frac{1}{\tau_{\text{sys}}} + f(I(t)) \right] \cdot x(t) + f(I(t)) \cdot A \quad (4)$$

This formulation has a striking structural similarity to the perfusion term in the BHTE ($Q_{\text{perfusion}} \propto -w_b(T - T_{\text{art}})$), which acts as a “leakage” mechanism. By employing LTCs, specifically structured via Neural Circuit Policies (NCPs), we aim to create a model that naturally learns the decay and diffusion dynamics of tissue heating, offering potentially superior generation to irregular sampling rates and improved stability.

Our proposed architecture, **Latent-LTC**, operates in a compressed latent space:

- 1) **Encoder:** A ResNet processes the frame V_t into a latent vector z_t .
- 2) **Dynamics:** An LTC layer evolves $z_t \rightarrow z_{t+1}$ using the learned ODE dynamics.
- 3) **Decoder:** A U-Net decoder reconstructs the dense temperature map $\hat{T}_{t+1}$ from the evolved state.

Implementation utilizes the `ncps` library, with experiments forthcoming in Phase 3.

F. Uncertainty Quantification

Reliability is paramount in clinical and industrial settings. We implement two strategies:

- **Deep Ensembles:** We train $M = 5$ independent instances of the SimpleResNet. The prediction mean and variance provide the estimate and aleatoric uncertainty.
- **Bayesian Neural Networks (BNN):** We employ Variational Inference (Bayes by Backprop [5]) to learn probability distributions over the network weights. We explore both a "Bayesian Head" (only final layers are Bayesian) and a "Full Bayesian" approach.

IV. IMPLEMENTATION AND BENCHMARKING

A. Training Infrastructure

The models were implemented in PyTorch. Training was parallelized using `tmux` sessions to maximize GPU utilization on an NVIDIA A100 cluster. Experiments were tracked and visualized using Weights & Biases (WandB).

B. Edge Deployment Simulation

To assess real-world feasibility, we developed a deployment pipeline using ONNX Runtime. We measured inference latency on a standard CPU (restricted to a single thread) to simulate the constraints of edge devices like the Raspberry Pi 4 or NVIDIA Jetson Nano.

V. RESULTS

A. Computational Efficiency

Table I presents the results of our edge benchmark. The "Total Latency" includes the 0.82 ms overhead for optical flow calculation.

TABLE I
EDGE DEVICE PERFORMANCE BENCHMARK

Model	Params (M)	GFLOPs	Total Latency (ms)	Simulated FPS
CNNLSTM	0.06	0.04	1.96	509
Pretrained	11.52	0.47	20.88	48
ResNet	11.45	0.16	8.84	113
Physics	0.13	0.08	18.53	54

Note: Latency measured on single-core CPU. Simulated FPS includes pre-processing time.

Our custom CNNLSTM achieves exceptional throughput (≥ 500 FPS), making it suitable for low-power microcontrollers. The ResNet-based models, while heavier, still comfortably achieve real-time performance (≥ 30 FPS) on the simulated hardware.

B. Predictive Performance

(Pending final training results from WandB)

VI. CONCLUSION

We demonstrated that physics-informed video regression is feasible on edge devices. By integrating optical flow and optimizing model architectures, we achieve robust temperature estimation with minimal latency. Future work will focus on deploying these models on physical hardware and validating the uncertainty estimates in real-world scenarios.

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